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Lower bounds on the eigenvalue gap for vibrating strings

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A B S T R A C T

We provide lower bounds on the eigenvalue gap for vibrating strings with fixed endpoints depending only on qualitative properties of the density function. For example, if the density ρ is symmetric on the interval $[0, a]$, and if λ_1 and λ_2 are the first two eigenvalues of $u''(x) + \lambda\rho(x)u(x) = 0$ in $(0, a)$ with $u(0) = u(a) = 0$ boundary conditions, then

$$\lambda_2 - \lambda_1 > \max\left\{\frac{1}{\int_0^{a/2}(\frac{a}{2} - x)\rho(x)\,dx}, \frac{\pi^2}{\rho_M a^2}\right\},$$

where $\rho_M = \max_{0 \leqslant x \leqslant a} \rho(x)$. The ideas used also lead to applications in the case of monotone densities.

1. Introduction

The frequencies $\omega = \sqrt{\lambda}$ of a vibrating string with fixed endpoints are determined by the eigenvalues of the Sturm–Liouville problem

$$\begin{cases} u''(x) + \lambda\rho(x)u(x) = 0 \quad \text{in } (0, a) \\ u(0) = u(a) = 0 \end{cases} \tag{1.1}$$

where $\rho(x)$ is a positive continuous function and represents the mass density. As is well known, the eigenvalues of (1.1) form a strictly increasing sequence of positive numbers which depend on the density $\rho(x)$. We denote them accordingly by

$$0 < \lambda_1[\rho] < \lambda_2[\rho] < \lambda_3[\rho] < \cdots.$$

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Bounds on ratios of eigenvalues, especially the ratio of the first two eigenvalues, have long been of interest (see e.g., [2,3,5,6] and references therein). Of related interest is the problem of obtaining optimal bounds on gaps of eigenvalues. Huang [7] has considered the gap of the first two eigenvalues of (1.1) for certain classes of symmetric densities and established several comparison results in the case of single-well densities and in the case of double-well densities. His approach is based on a variational principle for the gap.

The purpose of this paper is to provide lower bounds on the eigenvalue gap for (1.1). Here we will present a very elementary device which is also quite useful. The function $\varphi(x) = u_2(x)/u_1(x)$, where $u_1(x)$ and $u_2(x)$ are the first two eigenfunctions of (1.1), will play a key role in our discussion. Our lower bounds depend only on qualitative properties of the density function. For example, if the density ρ is symmetric on the interval $[0, a]$, then

$$\lambda_2[\rho] - \lambda_1[\rho] > \max\left\{\frac{1}{\int_0^{a/2}(\frac{a}{2} - x)\rho(x)\,dx}, \frac{\pi^2}{\rho_M a^2}\right\},$$

where $\rho_M = \max_{0\leqslant x\leqslant a}\rho(x)$. We also apply our lower bound techniques to strings with monotone densities.

An example is included in the last section, where we consider the eigenvalue problem of vibrating strings in the interval $(-a, a)$ with density $\rho(x) = |x|^m$, $m \geqslant 0$. We shall solve this problem by means of Bessel functions. We shall also compare our lower bound results with the actual eigenvalue gap.

2. Preliminaries

Let $u_n(x)$ be the nth eigenfunction of (1.1) corresponding to $\lambda_n[\rho]$, normalized so that

$$\int_0^a \rho(x)u_n^2(x)\,dx = 1.$$

It is known that $u_n(x)$ has exactly $(n-1)$ zeros in the open interval $(0, a)$. We may assume that $u_1(x) > 0$ on $(0, a)$, and that $u_2(x) > 0$ on $(0, x_0)$ and $u_2(x) < 0$ on (x_0, a), where $u_2(x_0) = 0$. Since $u_1''(x) = -\lambda_1[\rho]\rho(x)u_1(x) < 0$ on $(0, a)$, $u_1(x)$ is strictly concave on $(0, a)$. It follows that $u_1(x)$ has a unique critical point in $(0, a)$ so that for some $c_0 \in (0, a)$,

$$u_1'(x) > 0 \quad \text{on } (0, c_0) \quad \text{and} \quad u_1'(x) < 0 \quad \text{on } (c_0, a). \tag{2.1}$$

Let

$$\Gamma[\rho] = \lambda_2[\rho] - \lambda_1[\rho]$$

be the gap of the first two eigenvalues of (1.1), and define

$$\varphi(x) = \frac{u_2(x)}{u_1(x)}.$$

It was shown in [6] that $\varphi(x)$ is strictly decreasing on $(0, a)$. Also, by the differential equation and l'Hôpital's rule, we have

$$\varphi'(0) = \varphi'(a) = 0 \tag{2.2}$$

$$\varphi''(x) = -\Gamma[\rho]\rho(x)\varphi(x) - 2\varphi'(x)(\log u_1)'(x). \tag{2.3}$$

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We note that if the density ρ is symmetric on the interval $[0,a]$, then u_1 is symmetric while u_2 is antisymmetric on $[0,a]$, so that $x_0 = c_0 = a/2$.

A density function ρ is said to be left-balanced on the interval $[0,a]$ provided $\rho(x) \geqslant \rho(a-x)$ for all $x \in [0,a/2]$. It is said to be right-balanced on $[0,a]$ provided $\rho(x) \leqslant \rho(a-x)$ for all $x \in [0,a/2]$. We remark here that a monotone decreasing function is left-balanced on the interval $[0,a]$, and that a monotone increasing function is right-balanced on the interval $[0,a]$.

It is known that if ρ is left-balanced on $[0,a]$, then $x_0 \leqslant a/2$ and $c_0 \leqslant a/2$; while, if ρ is right-balanced on $[0,a]$, then $x_0 \geqslant a/2$ and $c_0 \geqslant a/2$ (see [1]).

3. Results

In this section and the next we shall describe methods for finding lower bounds on the eigenvalue gap. The properties of eigenfunctions are used throughout, but to obtain better results some information about the zero of $u_2(x)$ is needed.

We introduce here a function G on the interval $[0,a]$ associated with (1.1) by

$$G(x) = \varphi'(x) + \Gamma[\rho]\int_0^x \rho(t)\varphi(t)\,dt. \tag{3.1}$$

Then, by (2.2) and (2.3), we have $G(0) = 0$ and

$$G'(x) = -2\varphi'(x)\frac{u_1'(x)}{u_1(x)}.$$

As noted in Section 2, $\varphi(x)$ is strictly decreasing on $(0,a)$. Hence, by (2.1), $G'(x) > 0$ on $(0,c_0)$. It follows that $G(x) > 0$ on $(0,c_0)$, and then (3.1) gives

$$-\varphi'(x) < \Gamma[\rho]\int_0^x \rho(t)\varphi(t)\,dt \quad \text{on } (0,c_0). \tag{3.2}$$

There are now two cases to consider. First, if $c_0 \geqslant x_0$, then, integrating (3.2) over $[0,x_0]$ and using the fact that $\varphi(x_0) = 0$, we have

$$\begin{aligned}\varphi(0) &< \Gamma[\rho]\int_0^{x_0}\int_0^x \rho(t)\varphi(t)\,dt\,dx\\ &< \Gamma[\rho]\varphi(0)\int_0^{x_0}\int_0^x \rho(t)\,dt\,dx.\end{aligned}$$

From this it follows that

$$\Gamma[\rho] > \frac{1}{\int_0^{x_0}\int_0^x \rho(t)\,dt\,dx} = \frac{1}{\int_0^{x_0}(x_0-t)\rho(t)\,dt}. \tag{3.3}$$

Secondly, if $c_0 < x_0$, we consider the eigenvalue problem

$$\begin{cases} u''(x) + \lambda\tilde{\rho}(x)u(x) = 0 & \text{in } (0,a)\\ u(0) = u(a) = 0 \end{cases} \tag{3.4}$$

where $\tilde{\rho}(x) = \rho(a - x)$. Then the nth eigenfunction of (3.4) corresponding to $\lambda_n[\tilde{\rho}]$ is given by $\tilde{u}_n(x) = u_n(a - x)$. Thus, if $\tilde{c}_0$ is the critical point of $\tilde{u}_1(x)$, and if $\tilde{x}_0$ is the zero of $\tilde{u}_2(x)$ in $(0, a)$, then we have $\tilde{c}_0 = a - c_0$ and $\tilde{x}_0 = a - x_0$. Since $\tilde{c}_0 > \tilde{x}_0$, it follows from what we have just proved that

$$\Gamma[\tilde{\rho}] > \frac{1}{\int_0^{a-x_0} \int_0^x \rho(a-t)\,dt\,dx}.$$

But $\lambda_n[\tilde{\rho}] = \lambda_n[\rho]$ for all n, and thus, we obtain

$$\Gamma[\rho] > \frac{1}{\int_0^{a-x_0} \int_0^x \rho(a-t)\,dt\,dx} = \frac{1}{\int_{x_0}^a (t - x_0)\rho(t)\,dt}. \tag{3.5}$$

We have therefore proved:

Theorem 3.1. *If x_0 is the zero of $u_2(x)$ in $(0, a)$, then*

$$\Gamma[\rho] > \min\left\{\frac{1}{\int_0^{x_0} (x_0 - t)\rho(t)\,dt}, \frac{1}{\int_{x_0}^a (t - x_0)\rho(t)\,dt}\right\}.$$

Corollary 3.2. *If ρ is symmetric on $[0, a]$, then*

$$\Gamma[\rho] > \frac{1}{\int_0^{a/2} (\frac{a}{2} - t)\rho(t)\,dt}.$$

Proof. The condition on ρ implies that $x_0 = a/2$. □

Corollary 3.3. *If ρ is left-balanced on $[0, a]$, then*

$$\Gamma[\rho] > \min\left\{\frac{1}{\int_0^{a/2} (\frac{a}{2} - t)\rho(t)\,dt}, \frac{1}{\int_0^a t\rho(t)\,dt}\right\}.$$

A similar result holds for right-balanced densities.

Proof. As mentioned in Section 2, if ρ is left-balanced on $[0, a]$, then $x_0 \leqslant a/2$, and the corollary follows from Theorem 3.1. □

4. Further results

In this section some lower bound results are obtained by considering a different type of G-function.

Theorem 4.1. *If ρ is symmetric on $[0, a]$, then*

$$\Gamma[\rho] > \frac{\pi^2}{\rho_M a^2},$$

where $\rho_M = \max_{0 \leqslant x \leqslant a} \rho(x)$.

Proof. We define a function G on $[0, a]$ by

$$G(x) = \left[\varphi'(x)\right]^2 - \Gamma[\rho]\rho_M\left[\varphi^2(0) - \varphi^2(x)\right]. \tag{4.1}$$

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Since ρ is symmetric on $[0,a]$, φ is antisymmetric on $[0,a]$. Thus, G is symmetric on $[0,a]$, and by (2.2), we have $G(0)=G(a)=0$. We shall prove that $G(x)<0$ on $(0,a)$ by showing that $G'(x)<0$ on $(0,a/2)$. We compute

$$\begin{aligned} G'(x) &= 2\varphi'(x)\varphi''(x) + 2\Gamma[\rho]\rho_M\varphi(x)\varphi'(x) \\ &= 2\varphi'(x)\left\{\Gamma[\rho]\big[\rho_M-\rho(x)\big]\varphi(x) - 2\varphi'(x)\frac{u_1'(x)}{u_1(x)}\right\} \end{aligned}$$

by (2.3). Since $\varphi(x)$ is strictly decreasing and since $\varphi(x)$ and $u_1'(x)$ are positive on $(0,a/2)$, we see that $G'(x)<0$ on $(0,a/2)$. Hence $G(x)<0$ on $(0,a)$, or equivalently,

$$\sqrt{\Gamma[\rho]\rho_M} > \frac{-\varphi'(x)}{\sqrt{\varphi^2(0)-\varphi^2(x)}} \quad \text{on } (0,a).$$

Integrating from 0 to a, we get

$$a\sqrt{\Gamma[\rho]\rho_M} > \sin^{-1}\left[\frac{\varphi(x)}{\varphi(0)}\right]\bigg|_a^0 = \pi.$$

This proves the theorem. □

Theorem 4.2. *If ρ is symmetric on $[0,a]$ and if $\rho'(x)\geqslant 0$ on $(0,a/2)$, then*

$$\Gamma[\rho] > \frac{\pi^2}{(\int_0^a \sqrt{\rho(x)}\,dx)^2}.$$

Proof. Instead of (4.1) we now consider the G-function:

$$G(x) = \big[\varphi'(x)\big]^2 - \Gamma[\rho]\rho(x)\big[\varphi^2(0)-\varphi^2(x)\big].$$

Because of the symmetry of $\rho(x)$, we see that G is symmetric on $[0,a]$ with $G(0)=G(a)=0$. We compute

$$\begin{aligned} G'(x) &= 2\varphi'(x)\varphi''(x) + 2\Gamma[\rho]\rho(x)\varphi(x)\varphi'(x) - \Gamma[\rho]\rho'(x)\big[\varphi^2(0)-\varphi^2(x)\big] \\ &= -4\big[\varphi'(x)\big]^2\frac{u_1'(x)}{u_1(x)} - \Gamma[\rho]\rho'(x)\big[\varphi^2(0)-\varphi^2(x)\big] \end{aligned} \tag{4.2}$$

by (2.3). Using the hypothesis of $\rho'(x)$ and the fact that $u_1'(x)$ is positive on $(0,a/2)$, we see that $G'(x)<0$ on $(0,a/2)$. This implies that $G(x)<0$ on $(0,a)$. Thus, we have

$$\sqrt{\Gamma[\rho]\rho(x)} > \frac{-\varphi'(x)}{\sqrt{\varphi^2(0)-\varphi^2(x)}} \quad \text{on } (0,a).$$

Integrating from 0 to a, we obtain

$$\sqrt{\Gamma[\rho]}\int_0^a \sqrt{\rho(x)}\,dx > \pi,$$

and the theorem follows. □

We remark that by using variational principle arguments, it was established in [7] that if ρ is a symmetric single-barrier density on $[0,a]$, then

$$\Gamma[\rho] \geqslant \frac{3\pi^2}{\rho(a/2)a^2}$$

with equality only if ρ is a constant. In fact, the result remains true if ρ is a (not necessarily symmetric) single-barrier density on $[0,a]$ with transition point $x = a/2$ (see [7]). Thus, under the hypotheses of Theorem 4.2, we have

$$\Gamma[\rho] > \max\left\{\frac{\pi^2}{(\int_0^a \sqrt{\rho(x)}\,dx)^2}, \frac{3\pi^2}{\rho(a/2)a^2}\right\}.$$

Finally, we apply our lower bound techniques to strings with monotone densities.

Theorem 4.3. *If $\rho'(x) > 0$ on $(0,a)$ and if $\rho(x)/\rho'(x)$ is monotone increasing on $(a/2,a)$, then*

$$\Gamma[\rho] > \frac{\pi^2}{4(\int_0^a \sqrt{\rho(x)}\,dx)^2}.$$

Proof. We consider here the G-function:

$$G(x) = \left[\varphi'(x)\right]^2 - \Gamma[\rho]\rho(x)\left[b^2 - \varphi^2(x)\right], \tag{4.3}$$

where $b = \max\{\varphi(0), -\varphi(a)\}$. Since $\varphi(x)$ is strictly decreasing on $(0,a)$, we have $b^2 - \varphi^2(x) \geqslant 0$ for all $x \in [0,a]$. Also, by (2.2), we have

$$\begin{aligned} G(0) &= -\Gamma[\rho]\rho(0)\left[b^2 - \varphi^2(0)\right] \leqslant 0 \\ G(a) &= -\Gamma[\rho]\rho(a)\left[b^2 - \varphi^2(a)\right] \leqslant 0. \end{aligned} \tag{4.4}$$

From (2.3), we obtain, in the same way as we did (4.2) the result that

$$G'(x) = -4\left[\varphi'(x)\right]^2 \frac{u_1'(x)}{u_1(x)} - \Gamma[\rho]\rho'(x)\left[b^2 - \varphi^2(x)\right]. \tag{4.5}$$

Since $\rho'(x) > 0$ on $(0,a)$, (2.1) gives $G'(x) < 0$ on $(0,c_0)$, where $c_0 \geqslant a/2$, and it then follows from (4.4) that $G(x) < 0$ on $(0,c_0)$.

Our object is to prove that $G(x) \leqslant 0$ on $[0,a]$. Suppose on the contrary that $G(t) > 0$ for some $t \in (c_0,a)$. Then there must exist two points t_1 and t_2 in (c_0,a), where $t_1 < t_2$, such that

$$G'(t_1) = G'(t_2) = 0, \qquad G(t_1) < 0 \quad \text{and} \quad G(t_2) > 0.$$

From (4.3) and (4.5), it follows that

$$G(t_i) = \left[\varphi'(t_i)\right]^2 \left\{1 + 4\frac{u_1'(t_i)}{u_1(t_i)}\frac{\rho(t_i)}{\rho'(t_i)}\right\}$$

for $i = 1,2$. Since $G(t_1) < 0$ and $G(t_2) > 0$, this gives

$$\frac{u_1'(t_1)}{u_1(t_1)}\frac{\rho(t_1)}{\rho'(t_1)} < -\frac{1}{4} < \frac{u_1'(t_2)}{u_1(t_2)}\frac{\rho(t_2)}{\rho'(t_2)}. \tag{4.6}$$

As mentioned in Section 2, u_1 is strictly concave on $(0,a)$ so that u_1'/u_1 is strictly decreasing on $(0,a)$. Thus, using the hypothesis of ρ/ρ' and the fact that u_1'/u_1 is negative on (c_0,a), we obtain

$$\frac{u_1'(t_2)}{u_1(t_2)}\frac{\rho(t_2)}{\rho'(t_2)} \leqslant \frac{u_1'(t_2)}{u_1(t_2)}\frac{\rho(t_1)}{\rho'(t_1)} \leqslant \frac{u_1'(t_1)}{u_1(t_1)}\frac{\rho(t_1)}{\rho'(t_1)},$$

which contradicts (4.6). This proves that $G(x) \leqslant 0$ on $[0,a]$.

Now (4.3) gives

$$\sqrt{\Gamma[\rho]\rho(x)} \geqslant \frac{-\varphi'(x)}{\sqrt{b^2-\varphi^2(x)}} \quad \text{on } (0,a).$$

Integrating over $[0,a]$, we obtain

$$\sqrt{\Gamma[\rho]}\int_0^a \sqrt{\rho(x)}\,dx > \sin^{-1}\left[\frac{\varphi(0)}{b}\right] - \sin^{-1}\left[\frac{\varphi(a)}{b}\right] > \frac{\pi}{2}$$

and the theorem follows. □

5. A solvable example

The function known as Bessel function of the first kind of order n ($n \in \mathbb{R}$), and denoted by $J_n(x)$, can be defined by the infinite series

$$J_n(x) = \frac{x^n}{2^n\Gamma(n+1)}\left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2\cdot 4(2n+2)(2n+4)} - \cdots\right] \tag{5.1}$$

where Γ denotes the Gamma function.

In particular, by putting $n = 1/2$, $n = -1/2$, we find

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}}\sin x, \qquad J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}}\cos x.$$

Here we consider the eigenvalue problem of vibrating strings:

$$\begin{cases} u''(x) + \lambda\rho(x)u(x) = 0 & \text{in } (-a,a) \\ u(-a) = u(a) = 0 \end{cases} \tag{5.2}$$

where $\rho(x) = |x|^m$, $m \geqslant 0$.

It is known that the general solution of the equation

$$u''(x) + \lambda x^m u(x) = 0, \quad x \geqslant 0$$

is given by

$$u(x) = \sqrt{x}\left\{AJ_{\frac{1}{m+2}}\left(\frac{2\sqrt{\lambda}}{m+2}x^{\frac{m+2}{2}}\right) + BJ_{\frac{-1}{m+2}}\left(\frac{2\sqrt{\lambda}}{m+2}x^{\frac{m+2}{2}}\right)\right\}$$

where A and B are arbitrary constants [4, p. 118]. If the solution satisfies the initial condition $u(0) = 0$, we find by expanding the Bessel functions (5.1) that $B = 0$. On the other hand, if the solution satisfies the initial condition $u'(0) = 0$, we find that $A = 0$.

Now, since the density ρ is symmetric about the origin, it follows that the first two eigenfunctions of (5.2) are of the forms:

$$u_1(x) = B\sqrt{x}J_{\frac{-1}{m+2}}\left(\frac{2\sqrt{\lambda_1}}{m+2}x^{\frac{m+2}{2}}\right)$$
$$u_2(x) = A\sqrt{x}J_{\frac{1}{m+2}}\left(\frac{2\sqrt{\lambda_2}}{m+2}x^{\frac{m+2}{2}}\right)$$

for $x \in [0,a]$. Using the boundary conditions $u_1(a) = u_2(a) = 0$, we then obtain

$$J_{\frac{-1}{m+2}}\left(\frac{2\sqrt{\lambda_1}}{m+2}a^{\frac{m+2}{2}}\right) = J_{\frac{1}{m+2}}\left(\frac{2\sqrt{\lambda_2}}{m+2}a^{\frac{m+2}{2}}\right) = 0.$$

Thus, writing j_n for the least positive root of the equation $J_n(x) = 0$, we find that

$$\lambda_1 = \frac{(m+2)^2}{4a^{m+2}}j^2_{\frac{-1}{m+2}}, \qquad \lambda_2 = \frac{(m+2)^2}{4a^{m+2}}j^2_{\frac{1}{m+2}} \tag{5.3}$$

and hence

$$\Gamma\big[|x|^m\big] = \frac{(m+2)^2}{4a^{m+2}}\big(j^2_{\frac{1}{m+2}} - j^2_{\frac{-1}{m+2}}\big), \tag{5.4}$$

which is the actual eigenvalue gap for the density $\rho(x) = |x|^m$ on $[-a,a]$.

We shall now compare (5.4) with our lower bound results. From Corollary 3.2 and Theorem 4.1, we have

$$\Gamma\big[|x|^m\big] > \max\left\{\frac{1}{\int_0^a t\cdot t^m dt}, \frac{\pi^2}{a^m(2a)^2}\right\} = \begin{cases} \frac{\pi^2}{4a^{m+2}} & \text{if } 0 \leqslant m \leqslant (\pi^2/4) - 2 \\ \frac{m+2}{a^{m+2}} & \text{if } m > (\pi^2/4) - 2. \end{cases} \tag{5.5}$$

It is a classical result that j_n is an increasing function of n, if $n > -1$ [4, p. 107]. Thus, $j^2_{1/(m+2)} - j^2_{-1/(m+2)}$ decreases from $j^2_{1/2} - j^2_{-1/2} (= 3\pi^2/4)$ to 0 as m increases from 0 to ∞. Consequently, by (5.4) and (5.5), when m is large enough, the difference

$$\Gamma\big[|x|^m\big] - \frac{m+2}{a^{m+2}} = \frac{(m+2)^2}{4a^{m+2}}\left(j^2_{\frac{1}{m+2}} - j^2_{\frac{-1}{m+2}} - \frac{4}{m+2}\right)$$

is as small as we please, provided that $a > 1$. For small values of m, we can also make the difference

$$\Gamma\big[|x|^m\big] - \frac{\pi^2}{4a^{m+2}} = \frac{(m+2)^2}{4a^{m+2}}\left(j^2_{\frac{1}{m+2}} - j^2_{\frac{-1}{m+2}} - \frac{\pi^2}{(m+2)^2}\right)$$

as small as we please when a is sufficiently large.

Here are some numerical results for $f(m) = j^2_{1/(m+2)} - j^2_{-1/(m+2)} - 4/(m+2)$ and $g(m) = j^2_{1/(m+2)} - j^2_{-1/(m+2)} - \pi^2/(m+2)^2$:

m	1	10	10^2	10^3	10^4
$f(m)$	3.60841	0.90338	0.10629	0.01082	0.00108

m	0	0.1	0.2	0.3	0.4
$g(m)$	4.93480	4.81339	4.69308	4.57499	4.45983

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Finally, we would like to make a few remarks.

(1) A feature of the two lower bounds for $\Gamma[|x|^m]$ appearing in (5.5) is that for every fixed value of m, they become more accurate as a tends to infinity.

(2) From (5.3), we obtain the ratio of the first two eigenvalues:

$$\frac{\lambda_2}{\lambda_1} = \left(\frac{j_{\frac{1}{m+2}}}{j_{\frac{-1}{m+2}}}\right)^2.$$

The ratio is independent of a. Further, as m increases from 0 to ∞, it decreases from 4 to 1.

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